

the consumer segment through its BSNL collaboration. Last year, BSNL and Viasat trialled India's first D2D satellite connectivity service. That is, satellite communication using common devices like smartphones without any specialised hardware. At the India Mobile Congress, Viasat engineers demonstrated two-way and emergency SOS messaging with normal Android smartphones equipped for non-terrestrial network (NTN) connectivity. Messages were transmitted nearly 36,000km via one of Viasat's L-band geostationary satellites. Starting with messaging, they plan to slowly expand to full D2D internet services.

In a June 2025 media story, the company reaffirmed its commitment to expanding D2D services in India. While high-end phones like the Google Pixel can directly communicate with satellites, others will need a small 'puck' (a tiny device priced around ₹8000) to enable satellite connectivity.

Jio with SES. The joint venture JioSpaceFiber has successfully tested its satellite-based gigabit broadband service in remote locations like Gir (Gujarat), Korba (Chhattisgarh), Nabarangpur (Odisha), and the ONGC Plant in Jorhat (Assam). JioSpaceFiber is expected to complement Jio's existing fibre and 5G networks.

Eutelsat (OneWeb) with Bharti Airtel and Hughes Communications. OneWeb has already launched more than 70 satellites over India, using ISRO facilities. It has also set up feeder-link earth station gateways in Gujarat and Tamil Nadu to facilitate the delivery of satellite broadband services across India. Airtel and OneWeb have done successful proof-of-concepts in the Himalayas and underserved desert regions.

Starlink with... well, nobody and everybody. According to

telecom experts, Starlink's satellite capacity appears to be 80 to 90 times greater than competitors in India, like Airtel-OneWeb and Jio-SES. It plans to utilise its capacity efficiently using a multi-pronged, hybrid approach, where it will offer its internet services directly to consumers through its website, as well as use channel partners to penetrate the market quickly.

It has already partnered with Airtel and Jio for the sales and service of its equipment kits and subscription packs to consumers. It is also in talks with existing very small aperture terminal (VSAT) players, such as Hughes Communications, Nelco, and Viasat, to augment their multi-orbit satellite systems and offer hybrid connectivity solutions for the enterprise and government segments.

Can rural consumers afford satellite broadband?

Starlink aims to deliver internet connections with speeds ranging from 50Mbps to 250Mbps. With a total bandwidth capacity of approximately 600 to 700Gbps, it hopes to service thousands of simultaneous connections from across the country, predominantly rural and underserved areas.

To start using Starlink's satellite internet, users will need a starter kit that includes a dish antenna, mounting stand, Wi-Fi router, and necessary adaptors and cables. The standard kit is likely to be priced around ₹33,000. Starlink has announced that it intends to provide a month's free access to those who have purchased the kit, allowing them to experience it before deciding on their subscription plan, which is likely to cost between ₹3000 and ₹4200 per month. In many other countries, Starlink also offers a roaming/mobile kit, which

can be carried along when you travel. In the US and New Zealand, they also have a D2D service, limited to messaging. They have not yet announced these in India. We only have price estimates for the standard kit.

Although other players are yet to share their pricing estimates, it is likely to be comparable to Starlink's, especially in the initial stages. While satellite internet is probably the only option to connect certain rural and underserved areas digitally, we cannot help but wonder whether the common folks can afford it at such price points. Of course, it would be a boon for enterprises, estate owners, travellers, explorers, and others who can afford it. Remote villages can probably set up satellite broadband in community centres and share the setup and subscription costs, so everyone can benefit from it.

Government subsidies can help bring down the prices, especially in rural and unserved/underserved areas. Satellite internet companies can also attempt to reduce costs by means such as efficient spectrum use, sharing infrastructure with partners, and collaborating with existing terrestrial ISPs for backhaul or distribution. Technological advances, such as satellite miniaturisation, high-throughput satellites (HTS), reusable launch vehicles and rockets, and the mass production of standardised satellites, are helping to bring down the costs of setting up space infrastructure, which in turn can lower the final price points. Over time, growing competition will also bring down the costs and improve service quality for end users. Basically, we will see the laws of economics play out in this emerging market as soon as the government allocates the spectrum. Lights out, and away we go! **EFY**